

Shane Sarosh

shanesarosh@gmail.com | github.com/Shane12344321 | Kochi, India

EDUCATION

Vellore Institute of Technology (VIT)

Bachelor of Technology in Computer Science and Engineering; CGPA: 8.79/10

Vellore, India

Aug. 2024 – May 2028

EXPERIENCE

South Indian Bank

AI Intern

Kochi, India

May 2026 – Jun. 2026

- Built NoteForge, an AI-assisted QA platform that converts banking business-requirement documents into structured, traceable test cases using FastAPI, Pydantic, and LLM-based generation.
- Validated the system against digital-banking workflows spanning national eKYC identity verification (Aadhaar), core-banking integration (Finacle), and real-time payment rails (UPI/IMPS/NEFT/RTGS), including transaction limits and failure handling.

RESEARCH & PROJECTS

Searchless Chess: Mechanistic Interpretability | Python, JAX, Scikit-learn

[GitHub](#) | Jun. 2026

- Probed eight chess concepts across all 17 layers of DeepMind's 270M-parameter Searchless Chess transformer, training 136 linear probes on 5,488 positions.
- Recovered board state with near-perfect fidelity from layer activations (material balance $R^2 = 0.999$, center control $R^2 = 0.989$, near-ceiling castling-rights AUC).
- Found that this interpretable structure peaks in layers 1–3 before degrading as later layers shift to encoding action values.
- Built a Chrome extension and WebSocket pipeline that visualizes the transformer's move preferences in real time during post-game analysis.

Hybrid Job Hunter | Python, Playwright, GitHub Actions, Telegram API

[GitHub](#) | Jul. 2026

- Engineered adapters for seven ATS and career platforms, combining structured APIs with Playwright browser automation to monitor JavaScript-rendered career pages on scheduled GitHub Actions runs with no dedicated infrastructure.
- Designed for fault tolerance — per-source health tracking, request budgets, and queued retries — running unattended on a scheduled cadence.

Adaptive Shortest-Path Framework (ASPF) | C++, Python, Scikit-learn, NetworkX

Feb. 2026

- Developed a runtime system that dynamically selects among BFS, Dijkstra's algorithm, and Dial's algorithm by analyzing graph structural properties.
- Trained a Random Forest classifier that correctly selects the fastest shortest-path algorithm on 88% of held-out test graphs.
- Improved execution time by up to 73% over fixed Dijkstra baselines on sparse graphs and 58% on uniform-weight graphs.

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, $\text{\LaTeX}$

ML & Interpretability: JAX, Scikit-learn, linear probing, NetworkX

Tools & Infra: FastAPI, Pydantic, Playwright, GitHub Actions, Git, WebSockets, Chrome Extensions, Telegram API

CERTIFICATIONS

Microsoft Certified: Azure AI Fundamentals | Microsoft

[Credential](#)

Qiskit Global Summer School 2026 - Quantum Excellence | IBM

Aug. 2026